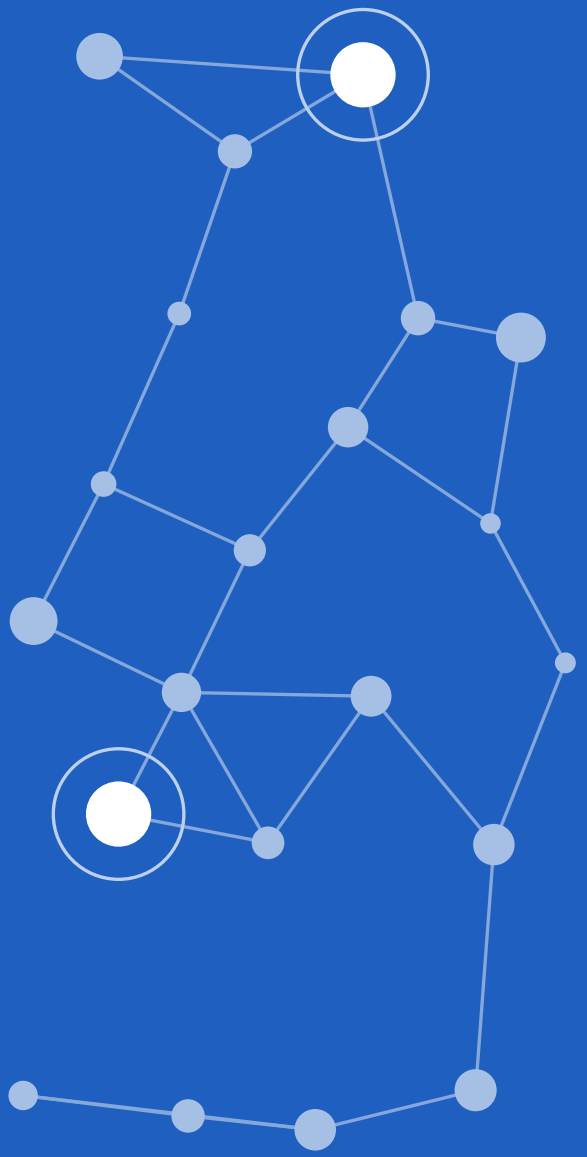
AI PROJECT ONBOARDING

Shared vocabulary turns AI ambition into sound project choices

Key concepts for understanding LLMs and AI agents



Strategy and engineering enablement · October 2026

THE WHOLE SYSTEM

A successful AI project designs the full system, not only the model

1	**Model capability**	Select a model and define what it can see in each interaction.
2	**Grounded knowledge**	Retrieve trusted, current enterprise information rather than relying on model memory alone.
3	**Controlled action**	Give the model narrow, observable tools instead of unrestricted system access.
4	**Measurement and control**	Evaluate quality, enforce guardrails, and monitor cost and latency in production.

Decision requested: Use this vocabulary to frame scope, owners, evidence, and trade-offs before selecting a solution.

Source: OpenAI — Tokens and tokenization; Lewis et al. — Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks; OpenAI — Function calling guide; NIST — AI Risk Management Framework

MODEL FUNDAMENTALS

Tokens and the context window set what the model can process in one interaction



Tokens are model-readable pieces of text

Text is split into tokens before processing; tokenization varies by model and language.



The context window is the model's working set

Instructions, conversation history, retrieved knowledge, and output all compete for the same finite window.



Context is a product design constraint

Decide what to include, summarize, retrieve, or omit so the model sees the relevant evidence.

Source: OpenAI — Tokens and tokenization

MODEL LIFECYCLE

Training builds a model; inference applies it to each request



Training changes model parameters

The learning process adjusts parameters from data to create general capabilities.



Inference applies the trained model

Each request combines an input and context with the existing model to produce an output.



The distinction changes project choices

Most enterprise work improves inference through prompts, retrieval, tools, and controls rather than training a model.

Source: Vaswani et al. — Attention Is All You Need ; OpenAI — Fine-tuning guide

GROUNDING

Embeddings make meaning searchable, and RAG uses that search to ground answers

Prepare knowledge	Create embeddings	Retrieve evidence	Generate with citations
			
Chunk approved documents and attach permissions, provenance, and update rules.	Encode chunks as vectors so related meanings are located near each other.	Match the request to relevant, authorized chunks at run time.	Give the model retrieved evidence and ask it to answer from that evidence.

Source: Lewis et al. — Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

ADAPTATION CHOICE

Prompting steers a model now, while fine-tuning changes behavior through training

	RECOMMENDED **Prompting** Instructions and examples at request time	**Fine-tuning** Additional training on representative examples
Best when	**The task or policy changes often**	Behavior needs a stable, repeatable pattern
Changes	**Input context and instructions**	Learned model behavior
Typical project work	**Design prompts, templates, retrieval**	Curate, label, split, and test a training set
Main governance need	**Version prompts and test regressions**	Manage data quality, privacy, and evaluation

Verdict: Start with prompting and retrieval; fine-tune only when evidence shows a persistent gap.

Source: OpenAI — Prompt engineering; OpenAI — Fine-tuning guide

RELIABILITY

Hallucinations are plausible but unsupported outputs, so systems need multiple reduction layers



Ground each answer in approved evidence

Retrieve trusted sources and require citations where users need to verify an answer.



Constrain important outputs

Use structured schemas, validation, and policy checks before a response or action is accepted.



Clarify or escalate ambiguity

Ask for missing context and route high-consequence cases to a qualified person.



Govern the knowledge base

Set provenance, access, freshness, and review rules for the sources used at run time.

Source: Lewis et al. — Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks ; OpenAI — Function calling guide ; NIST — AI Risk Management Framework

CONTROLLED ACTION

Tool use lets a model act, and function calling supplies the controlled contract

01



Declare allowed tools

Describe each operation with a name, purpose, and typed input schema.

02



Request a structured call

The model selects a tool and produces arguments that match the schema.

03



Validate and execute

The application checks authorization, policy, and inputs before acting.

04



Observe the result

Return a bounded result to the model and log the action for review.

Source: OpenAI — Function calling guide; NIST — AI Risk Management Framework

AI AGENTS

Agents repeat plan–act–observe loops, with memory treated as managed state

Plan



Interpret the goal, choose the next step, and define what success looks like.

Act



Generate a response or invoke an approved tool within explicit limits.

Observe



Read the tool result or user feedback and decide whether more work is needed.

Manage memory



Keep only useful state; separate short-lived context from durable, governed records.

Source: Anthropic — Building effective agents; NIST — AI Risk Management Framework

QUALITY AND SAFETY

Evaluation and guardrails make quality, safety, and operational control testable



Define success and failure

Build representative tasks, expected outcomes, and unacceptable failure modes into an evaluation set.



Test before release

Compare quality, safety, and tool-use behavior against agreed acceptance criteria.



Guard at run time

Apply access controls, content policies, schemas, approvals, and human escalation.



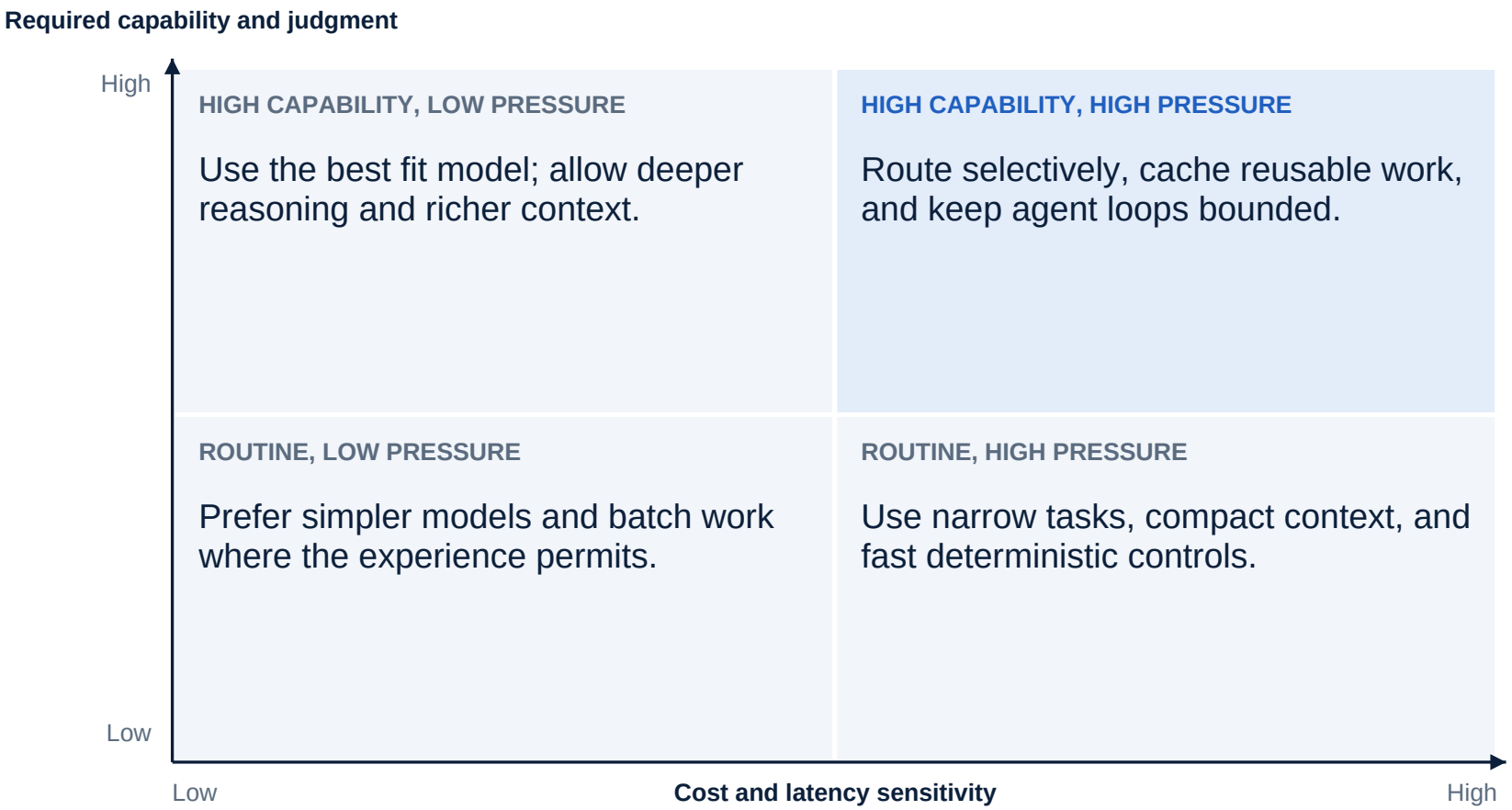
Monitor in production

Log outcomes, analyze failures, and refresh evaluations as the product and knowledge change.

Source: OpenAI — Evaluation best practices; NIST — AI Risk Management Framework

OPERATING TRADE-OFFS

Capability, cost, and latency form a design trade-off to optimize deliberately



Set a service-level target first; then choose a model, context strategy, and control loop to meet it.

Source: OpenAI — Latency optimization; OpenAI — Tokens and tokenization

FROM VOCABULARY TO CHARTER

A short discovery sequence turns the vocabulary into an AI project charter

Next step	Owner	Due
Select a narrow user problem and define the decision or outcome it should improve	Business sponsor	Discovery start
Specify the context, trusted knowledge, tools, and permissions the system may use	Product and engineering	Solution design
Define evaluation cases, failure thresholds, guardrails, and escalation paths	Product and risk owner	Before pilot
Set quality, cost, and latency targets before choosing the production architecture	Technology lead	Before build

Source: OpenAI — Evaluation best practices; NIST — AI Risk Management Framework; OpenAI — Latency optimization